

# HelixConstitution Gates — Lava Implementation Status

**Last updated:** 2026-06-14 (governance-debt audit 173e9747 re-verification: corrected stale rows — CM-SUBAGENT-DELEGATION-AUDIT scanner now exists + verify-all-wired; CM-COVERAGE-LEDGER is STRICT-default; CM-CHALLENGE-COVERAGE scanner STRICT-default; retired CM-LVA-TICKETS-SYNC → CM-WORKABLE-ITEMS-SYNC; build-stats tracker + script-docs-sync confirmed wired). Prior: 2026-05-16 (HelixQA Phase 4-debt closed — upstream PR b13ba7c landed helix-deps.yaml + install\_upstreams.sh; 17/17 own-org submodules satisfy CM-HELIX-DEPS-MANIFEST + CM-CANONICAL-ROOT-CLARITY + CM-INSTALL-UPSTREAMS-RAN in fully STRICT mode) **Inheritance:** Per Lava CLAUDE.md §6.AD (HelixConstitution Inheritance Mandate); enumerated per §6.AD-debt item 8.

## Purpose

HelixConstitution defines a set of CM-\* mechanical gates (see constitution/Constitution.md §11.4 sub-clauses). Each gate has a status: wired, paper-only, or blocked. This document is the operator-readable index — the source of truth for "what's actually enforced versus what's documented."

## Gate Inventory

| Gate ID                        | HelixConstitution clause           | Lava implementation                                                                                                                                                                                     | S |
|--------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| CM-COVENANT-114-16-PROPAGATION | §11.4.16 (Item-Type tracking)      | n/a (Lava uses .lava-ci-evidence/crashlytics-resolved/ + .lava-ci-evidence/sixth-law-incidents/ rather than Issues/Issues_Summary trackers; equivalence documented in §6.AD.3)                          | 0 |
| CM-COMMIT-DOCS-EXISTS          | §11.4.22 (lightweight commit path) | scripts/check-commit-docs-exists.sh (present + executable; verify-all-wired at scripts/verify-all-constitution-rules.sh:190 over LAVA_COMMIT_RANGE='HEAD~5..HEAD') — verifies docs referenced in recent |   |

| Gate ID                                   | HelixConstitution clause                                   | Lava implementation                                                                                                                                                                                                                                                                                                                                                         | S   |
|-------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| CM-FIXED-COLUMN-ALIGNMENT                 | §11.4.19 (Fixed-document column alignment)                 | commit messages resolve (2026-06-14 re-verification)<br>n/a (Lava does not maintain Fixed.md / Fixed_Summary.md tracker structure; functional equivalent: per-issue closure logs in .lava-ci-evidence/crashlytics-resolved/)                                                                                                                                                | 0.0 |
| CM-SCRIPT-DOCS-SYNC                       | §11.4.18 (script documentation mandate)                    | .github/pre-push Check 9 — fires when scripts/X.sh is modified without companion docs/scripts/X.sh.md in the same commit (only when the doc already exists at this commit); rehearsed in synthetic fixture                                                                                                                                                                  | 0.0 |
| CM-BUILD-RESOURCE-STATS-TRACKER           | §11.4.24 (build-resource stats tracking)                   | scripts/build-stats-sample.sh (per-sample collector, BSD+GNU portable, wraps a build via wrap "<name>" --<cmd>) + .lava-ci-evidence/build-stats/registry.tsv (per-build summary registry) + scripts/build-stats-report.sh (derives docs/build-stats/Stats.md + Stats.html via pandoc + Stats.pdf via wkhtmltopdf-or-weasyprint; HTML/PDF skip silently if exporters absent) | 0.0 |
| CM-ITEM-STATUS-TRACKING                   | §11.4.15 (item-status tracking)                            | n/a (Lava uses CONTINUATION.md + closure logs; mapping documented in §6.AD.3)                                                                                                                                                                                                                                                                                               | 0.0 |
| CM-ITEM-TYPE-TRACKING                     | §11.4.16 (item-type tracking)                              | n/a (same as above)                                                                                                                                                                                                                                                                                                                                                         | 0.0 |
| CM-ITEM-OPERATOR-BLOCKED-DETAILS          | §11.4.21 (operator-blocked status + self-resolution audit) | n/a (Lava uses .lava-ci-evidence/sixth-law-incidents/<date>.json for blocker tracking; the Operator-blocked status vocabulary is not in use)                                                                                                                                                                                                                                | 0.0 |
| CM-OPERATOR-BLOCKED-SELF-RESOLUTION-AUDIT | §11.4.21                                                   | n/a (same as above)                                                                                                                                                                                                                                                                                                                                                         | 0.0 |
| CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION    | §11.4.17 (universal-vs-project classification)             | .github/pre-push Check 8 — fires when ##### 6.X clause added to CLAUDE.md without Classification: line in commit body                                                                                                                                                                                                                                                       | 0.0 |
| CM-SUBAGENT-DELEGATION-AUDIT              | §11.4.20 (subagent-driven-by-default mandate)              | scripts/check-subagent-delegation-audit.sh (present + executable; verify-all-wired at scripts/verify-all-constitution-rules.sh:193 over LAVA_COMMIT_RANGE='HEAD~5..HEAD') —                                                                                                                                                                                                 | (S  |

| Gate ID                                               | HelixConstitution clause                                                                                                                                                                | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | S |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| CM-WORKABLE-ITEMS-SYNC (formerly CM-LVA-TICKETS-SYNC) | §11.4.93 (SQLite single-source-of-truth) + §11.4.95 (DB tracked-in-git, never gitignored) + §11.4.106 (byte-identical doc↔DB round-trip) + §11.4.33/34 (closure + reopen schema guards) | <p>scanner-level audit of the commit-range delegation trail (2026-06-14 re-verification; CLAUDE.md/this-doc previously marked paper-only)</p> <p><b>2026-06-14 correction:</b> the LVA-specific tools/lava-tickets/ system + scripts/check-lva-tickets.sh were RETIRED (per §6.AF, LVA-3 migrated to the canonical workable-items Go binary; the old script is ABSENT). Current gate: scripts/check-workable-items.sh (present + executable; verify-all-wired at scripts/verify-all-constitution-rules.sh:202) operates on docs/workable_items.db via the canonical constitution/scripts/workable-items binary (validate+diff+DB-tracked). Hermetic test: tests/check-workable-items/. Upstream update/reopen/block + full export pipeline OWED upstream (non-blocking).</p> <p>scripts/sync-markdown-exports.sh (--check-only/--regenerate-all/--regenerate &lt;f&gt;; pandoc HTML + pandoc --pdf-engine=weasyprint PDF; timeout 60/file; 500-candidate cap with log) + scripts/check-markdown-export-sync.sh (gate wrapper; LAVA_MARKDOWN_EXPORT_STRICT advisory/strict). Scope: project-root + docs/** + scripts/**.md; excludes source trees + submodules + constitution (they sync their own). Backfill 2026-05-31: 126 in-scope docs, generated .html+.pdf siblings (TRACKED per §11.4.65). Hermetic test: tests/check-markdown-export-sync/test_markdown_export_sync.sh (positive + delete-sibling-FAIL + backdate-mtime-FAIL + restore-PASS). Closes §6.AF-debt §11.4.65 item.</p> <p>scripts/check-non-fatal-coverage.sh (bash scanner; default STRICT-mode after queue drained 2026-05-14; LAVA_NONFATAL_STRICT=0 to revert to advisory mode)</p> |   |
| CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC                     | HelixConstitution §11.4.65 Universal Markdown Export Mandate                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| CM-NONFATAL-COVERAGE                                  | §6.AC + HelixConstitution telemetry discipline                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| Gate ID                          | HelixConstitution clause                                               | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | S                                                   |
|----------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| CM-CHALLENGE-DISCRIMINATION      | §6.AB Anti-Bluff Test-Suite Reinforcement                              | scripts/check-challenge-discrimination.sh (bash scanner; STRICT-default; verifies every Challenge*Test.kt carries a FALSIFIABILITY REHEARSAL marker in KDoc OR a companion .lava-ci-evidence/sp3a-challenges/.json; LAVA_CHALLENGE_DISCRIMINATION_STRICT=0 to revert to advisory)                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                     |
| CM-CHALLENGE-COVERAGE            | §6.AE Comprehensive Challenge Coverage + Container/QEMU Matrix Mandate | scripts/check-challenge-coverage.sh (bash scanner; STRICT-default after queue drained 2026-05-15; verifies every feature/* module has at least one Challenge that targets it via import / // covers-feature: marker / heuristic / // AE-exempt: ledger entry; LAVA_CHALLENGE_COVERAGE_STRICT=0 to revert to advisory) + scripts/run-challenge-matrix.sh (matrix-runner glue; delegates to submodules/containers/cmd/emulator-matrix --runner=containerized; pre-bakes the §6.AE.2 minimum API/form-factor matrix; refuses to claim success on hosts lacking KVM with a clear §6.X-debt host-gap diagnostic). Hermetic test: tests/check-constitution/test_challenge_coverage.sh (4 falsifiability fixtures, all PASS) | (I<br>sc<br>S<br>g<br>ru<br>B<br>d<br>a<br>st<br>§6 |
| CM-VERIFY-ALL-CONSTITUTION-RULES | HelixConstitution §11.4.32 Post-Constitution-Pull Validation Mandate   | scripts/verify-all-constitution-rules.sh (master sweep; STRICT default; wraps every individual gate above + every hermetic test suite into a single enforcement engine; emits per-run attestation JSON at .lava-ci-evidence/verify-all/<UTC-timestamp>.json with per-gate pass/fail/duration); wired into scripts/ci.sh step 5a4. Hermetic test: tests/check-constitution/test_verify_all_rules.sh (4 falsifiability fixtures including the §11.4.32-mandated paired-mutation that plants a known violation + asserts sweep reports FAIL). PER §11.4.32 ITSELF: this gate is "the enforcement engine for every other §11.4.x and CONST-NNN rule" —                                                                    | (I<br>cc<br>cc<br>p<br>4                            |

| Gate ID                                                         | HelixConstitution clause                                                     | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | S                        |
|-----------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| CM-GITIGNORE-<br>PRECOMMIT-AUDIT                                | HelixConstitution §11.4.30 .gitignore + No-Versioned-Build-Artifacts Mandate | <p>without it, new rules cascade as anchors but never get enforced in the codebase</p> <p>scripts/check-gitignore-coverage.sh (STRICT default; verifies every actual Gradle module + every owned-by-us submodule has .gitignore; verifies no tracked files match the forbidden-pattern set with explicit allowlist for documented-non-sensitive deployment configs); wrapped by verify-all sweep. Hermetic test: tests/check-constitution/test_gitignore_coverage.sh (3 fixtures: real-tree-passes, module-without-gitignore-rejected, clean-fixture-passes)</p> <p>scripts/check-no-nested-own-org-submodules.sh (STRICT default; walks every owned submodule's .gitmodules; rejects entries pointing to vasic-digital / HelixDevelopment / red-elf / ATMOSphere1234321 / Bear-Suite / BoatOS123456 / Helix-Flow / Helix-Track / Server-Factory orgs on github.com or gitlab.com; per-entry waiver mechanism with mandatory rationale + tracking-issue + target-removal-date); wrapped by verify-all sweep in STRICT mode. Hermetic test: tests/check-constitution/test_nested_own_org_submodules.sh (4 fixtures: clean-fixture-passes, vasic-digital-chain-rejected, HelixDevelopment-gitlab-chain-rejected, advisory-mode-returns-zero). Phase 5-debt closed 2026-05-15: github cascade merge removed Challenges/.gitmodules (CONST-051(C) flat-layout enforcement); Panoptic no longer nested; scanner reports 0 violations in STRICT mode</p> | (I<br>co<br>co<br>p      |
| CM-NO-NESTED-<br>OWN-ORG-<br>SUBMODULES                         | HelixConstitution §11.4.28 Submodules-As-Equal-Codebase Mandate              | <p>scripts/check-canonical-root-and-upstreams.sh (STRICT default; verifies (a) root CLAUDE.md + AGENTS.md open with ## INHERITED FROM constitution/&lt;file&gt; or @constitution/</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | S<br>(E<br>co<br>co<br>p |
| CM-CANONICAL-<br>ROOT-CLARITY +<br>CM-INSTALL-<br>UPSTREAMS-RAN | HelixConstitution §11.4.35 Canonical-Root Inheritance Clarity + §11.4.36     | <p>scripts/check-canonical-root-and-upstreams.sh (STRICT default; verifies (a) root CLAUDE.md + AGENTS.md open with ## INHERITED FROM constitution/&lt;file&gt; or @constitution/</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | S<br>(E<br>co<br>co<br>c |

| Gate ID                | HelixConstitution clause                                         | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | S                               |
|------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                        | Mandatory<br>install_upstreams                                   | <p>&lt;file&gt; pointer in first 40 lines, (b) constitution submodule's three canonical files exist, (c) constitution submodule's own CLAUDE.md/ AGENTS.md do NOT carry ## INHERITED FROM at top level — fenced ```markdown code blocks correctly recognized as documentation; verifies every owned submodule has install_upstreams script at one of 6 supported locations; per-submodule waiver mechanism with mandatory rationale); wrapped by verify-all sweep in STRICT mode. Hermetic test: tests/check-constitution/test_canonical_root_and_upstreams.sh (6 fixtures including discrimination test for fenced-code-block false-positive prevention). Phase 8-debt closed 2026-05-15: 10 install_upstreams scripts landed (Challenges, Config, Containers, Discovery, HTTP3, Mdns, Middleware, RateLimiter, Recovery, Tracker-SDK each gained install_upstreams.sh + Upstreams/GitHub.sh + Upstreams/GitLab.sh per §6.W). Phase 4-debt closed 2026-05-16: HelixQA upstream PR b13ba7c (HelixDevelopment/HelixQA) landed install_upstreams.sh wrapper at HelixQA's repo root + Lava parent removed HelixQA from INSTALL_UPSTREAMS_WAIVERS; scanner now reports 17/17 install_upstreams present (0 waived) + all §11.4.35 sub-checks passing in STRICT mode scripts/check-helix-deps-manifest.sh (STRICT default; verifies (a) parent project root has helix-deps. {yaml,json,toml} with schema_version: 1 + deps: + transitive_handling: blocks, (b) every owned-by-us submodule has its own helix-deps. {yaml,json,toml}); per-submodule waiver mechanism with mandatory rationale); wrapped by verify-all sweep in STRICT mode. Hermetic test: tests/check-constitution/</p> | p<br>4<br>2                     |
| CM-HELIX-DEPS-MANIFEST | HelixConstitution §11.4.31 Submodule-Dependency-Manifest Mandate | <p>{yaml,json,toml} with schema_version: 1 + deps: + transitive_handling: blocks, (b) every owned-by-us submodule has its own helix-deps. {yaml,json,toml}); per-submodule waiver mechanism with mandatory rationale); wrapped by verify-all sweep in STRICT mode. Hermetic test: tests/check-constitution/</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | S<br>(<br>c<br>c<br>p<br>4<br>2 |



| Gate ID            | HelixConstitution clause                                    | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | S                                   |
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| CM-COVERAGE-LEDGER | HelixConstitution §11.4.25 Full-Automation-Coverage Mandate | <p>test_helix_deps_manifest.sh (6 fixtures: compliant-fixture-passes, missing-parent-rejected, missing-submodule-rejected, wrong-schema-version-rejected, advisory-mode-returns-zero, json-variant-accepted). Phase 3-debt closed 2026-05-15: 16/16 per-submodule helix-deps.yaml manifests landed (Auth, Cache, Challenges, Concurrency, Config, Containers, Database, Discovery, HTTP3, Mdns, Middleware, Observability, RateLimiter, Recovery, Security, Tracker-SDK — each authored with honest per-submodule deps after analysis: 15 leaf modules with empty deps + Challenges declaring Panoptic with layout: flat). Phase 4-debt closed 2026-05-16: HelixQA upstream PR b13ba7c (HelixDevelopment/HelixQA) landed helix-deps.yaml at HelixQA's repo root (declares Challenges + Containers deps per HelixQA's README sibling-directory enumeration) + Lava parent removed HelixQA from HELIX_DEPS_WAIVERS; parent helix-deps.yaml declares 18 deps (16 vasic-digital + HelixDevelopment/HelixQA + constitution); scanner now reports 17/17 per-submodule manifests present (0 waived) in STRICT mode</p> <p>scripts/generate-coverage-ledger.sh (deterministic per-row YAML generator; walks feature/*/, core/*/, app/, lava-api-go/, submodules/*/; emits 6 invariants per row — anti_bluff, working_capability, documented_promise, no_open_bugs, documented, four_layer — with status pass/gap/n/a/waiver;; produces docs/coverage-ledger.yaml) + scripts/check-coverage-ledger.sh (verifier; asserts schema-presence, every-on-disk-path-has-a-row, AND ledger-fresh-vs-regenerated). Companion waiver mechanism: docs/coverage-ledger.waivers.yaml (hand-edited per-row overrides for invariants 3/4 the</p> | S<br>(I<br>co<br>co<br>p<br>S<br>cl |

| Gate ID                            | HelixConstitution clause                                                                                    | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | S |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| CM-CLOSURE-STATUS-VOCAB-COMPLIANCE | HelixConstitution §11.4.33 Type-aware closure-status vocabulary                                             | <p>generator cannot mechanically infer). Hermetic test: tests/check-constitution/test_coverage_ledger.sh (6 fixtures: clean-fixture-passes, missing-ledger-rejected, missing-row-rejected, stale-ledger-rejected, advisory-mode-returns-zero, generator-deterministic). <b>2026-06-14 correction:</b> STRICT flip is DONE — scripts/check-coverage-ledger.sh:41 STRICT=\$ {LAVA_COVERAGE_LEDGER_STRICT:-1} (strict default) and verify-all invokes it --strict (scripts/verify-all-constitution-rules.sh:156). Phase-7-debt STRICT-flip CLOSED.</p> <p>EQUIVALENCE-MAPPED per §6.AD.3 (Phase 9 Path B): Lava's .lava-ci-evidence/sixth-law-incidents/&lt;date&gt;-&lt;slug&gt;.json schema includes candidate-causes, ELIMINATED entries, and PENDING_FORENSICS: markers (per §11.4.6) that satisfy the type-aware closure-status semantics §11.4.33 prescribes. No standalone scanner — the §6.AD.3 mapping IS the binding equivalence per Path B.</p> <p>EQUIVALENCE-MAPPED per §6.AD.3 (Phase 9 Path B): Lava's .lava-ci-evidence/crashlytics-resolved/&lt;date&gt;-&lt;slug&gt;.md closure-log schema includes root-cause + fix-commit-SHA + validation-test references — providing WHY + WHO + WHEN + WHICH-INCIDENT lineage for each closed item. Re-opens are tracked as new closure-log entries citing the prior log. No standalone scanner — the §6.AD.3 mapping IS the binding equivalence per Path B.</p> |   |
| CM-REOPENED-SOURCE-ATTRIBUTION     | HelixConstitution §11.4.34 Reopened-source attribution                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| CM-HELIXQA-WIRING                  | HelixConstitution §11.4.27 (CONST-050) full test-type coverage including HelixQA + Lava §6.AE Comprehensive | scripts/run-helixqa-challenges.sh (Lava-side wrapper invoking the 11 HelixQA Challenge scripts at submodules/helixqa/challenges/scripts/; captures per-script stdout+stderr to log files; emits roll-up attestation JSON; classifies exit codes 0/2/other → PASS/SKIP/FAIL;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |



| Gate ID               | HelixConstitution clause                                                               | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | S                                                                        |
|-----------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|                       | Challenge Coverage                                                                     | <p>refuses to silently mask missing submodule or wiring drift — exit 3 with actionable remediation per §6.J anti-bluff posture). Wired into scripts/run-challenge-matrix.sh via --include-helixqa opt-in flag (OFF by default so existing matrix runs are unaffected). Phase 4 follow-up B (2026-05-16) added: --runner=host containerized flag (§6.X container delegation, honest-fail-fast exit 4 on missing Containers submodule), --require-toolchain=go none flag (§6.J toolchain precondition gate with HELIXQA_TOOLCHAIN_MAP per-script declaration),</p> <p>LAVA_HELIXQA_EVIDENCE_DIR env-var override (Q3), HELIXQA_W_EXCLUSIONS array consuming docs/helixqa-script-audit.md (Q4). Hermetic test: tests/check-constitution/test_helixqa_wiring.sh (11 fixtures covering all 4 questions' resolutions + the original 4 fixtures). Docs: docs/scripts/run-helixqa-challenges.sh.md + docs/scripts/run-challenge-matrix.sh.md + docs/helixqa-script-audit.md. Design: docs/plans/2026-05-16-helixqa-integration-design.md Option 1 — shell-level wiring. Many HelixQA scripts are Linux-only (systemctl/journalctl); macOS hosts will see FAIL/SKIP outcomes that are HONEST acknowledgments of the wrong-OS gap rather than false-pass.</p> <p>docs/helixqa-script-audit.md (per-script audit of all 11 HelixQA Challenge scripts; 4-query methodology: git push / curl-wget / external HTTP / outside-worktree FS writes; per-script verdict table with COMPLIANT / REQUIRES-EXCLUSION / HOST-MUTATION-RISK classification; aggregate verdict + operator-supplied env-var risk surface + re-audit triggers). Consumed by scripts/run-helixqa-challenges.sh via the</p> | <p>co<br/>ru<br/>fo<br/>g</p>                                            |
| CM-HELIXQA-§6.W-AUDIT | Lava §6.W (GitHub + GitLab Only Remote Mandate) + §6.J (Anti-Bluff Functional Reality) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>(a<br/>P<br/>a<br/>d<br/>m<br/>s<br/>O<br/>fu<br/>b<br/>in<br/>vi</p> |

| Gate ID | HelixConstitution clause | Lava implementation                                                                                                                                                                                                                                                                                                                                                                                  | S |
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|         |                          | HELIXQA_W_EXCLUSIONS array constant (currently empty since 0/11 scripts violate §6.W on default config). Re-audit OWED whenever the submodules/helixqa pin bumps; the wrapper's wiring-drift check (exit 3 on added/removed scripts) prompts re-audit but does not mechanize the per-script grep — that remains a human-driven audit per §6.J ("real grep results only — no manufactured findings"). |   |

## Lava-side anti-bluff gates (NOT HelixConstitution CM-\* but related)

For completeness — these gates predate or extend HelixConstitution's CM-\* set: 

| Gate name                                        | Lava clause           | Implementation                                                            | Status  |
|--------------------------------------------------|-----------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Hosted-CI config rejection                       | Local-Only CI/CD rule | .githubhooks/pre-push Check 1                                             | wired   |
| Test-commit Bluff-Audit stamp                    | Seventh Law clause 1  | .githubhooks/pre-push Check 2                                             | wired                                                                                        |
| Mock-the-SUT pattern rejection                   | Seventh Law clause 4  | .githubhooks/pre-push Check 3                                             | wired   |
| Gate-shaping file Bluff-Audit stamp + path-match | §6.N.1.2              | .githubhooks/pre-push Check 4 (with extended .githubhooks/pre-push regex) | wired   |
| Attestation falsifiability rehearsal             | §6.N.1.3              | .githubhooks/pre-push Check 5                                             | wired                                                                                        |
| §6.Y bump-first ordering                         | §6.Y                  | .githubhooks/pre-push Check 6                                             | wired   |
| §6.Z evidence-file presence on                   | §6.Z                  | .githubhooks/pre-push Check 7 + firebase-                                 | wired                                                                                        |

| Gate name                                                                             | Lava clause                               | Implementation                                                                                                                           | Status               |
|---------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| pointer<br>advance                                                                    |                                           | distribute.sh<br>Phase-1 Gates                                                                                                           | ☑                    |
| §6.AA Two-<br>Stage<br>default +<br>release-<br>requires-<br>debug-<br>companion      | §6.AA                                     | firebase-<br>distribute.sh<br>(default flipped<br>to MODE=debug; --<br>release-only<br>requires last-<br>version-debug >=<br>current vc) | ☑ wired              |
| §6.W mirror-<br>host<br>boundary                                                      | §6.W + §6.AD.1                            | scripts/check-<br>constitution.sh<br>6.AD §6.W block                                                                                     | ☑ wired              |
| §6.AD per-<br>scope<br>inheritance<br>pointer-<br>block<br>presence                   | §6.AD.4                                   | scripts/check-<br>constitution.sh<br>6.AD(4); 54 in-<br>scope files<br>enforced                                                          | ☑ wired              |
| §6.AD<br>constitution<br>submodule<br>presence +<br>structure                         | §6.AD.2                                   | scripts/check-<br>constitution.sh<br>6.AD(2)                                                                                             | ☑ wired              |
| §6.AD<br>commit_all.sh<br>+ inject-<br>helix-<br>inheritance-<br>block.sh<br>presence | §6.AD.5 +<br>§6.AD.6                      | scripts/check-<br>constitution.sh<br>6.AD(5)+(6)                                                                                         | ☑ wired              |
| §11.4.6 no-<br>guessing<br>vocabulary                                                 | §6.AD.6 +<br>HelixConstitution<br>§11.4.6 | scripts/check-<br>constitution.sh<br>final block                                                                                         | ☑ wired              |
| §6.X<br>Container-<br>Submodule<br>Emulator<br>wiring                                 | §6.X                                      | scripts/check-<br>constitution.sh<br>6.X(4)+(5) (with<br>PARTIAL CLOSE<br>— Linux x86_64<br>gate-host still<br>owed)                     | ☑ wired<br>(PARTIAL) |
| §6.K<br>Containers<br>extension<br>presence                                           | §6.K                                      | scripts/check-<br>constitution.sh<br>clause 6.K block                                                                                    | wired                |

# Why some gates are paper-only

The CM-\* gates marked paper-only typically depend on infrastructure Lava does not have today:

- **Issues / Issues\_Summary / Fixed / Fixed\_Summary trackers** (§11.4.15 / §11.4.16 / §11.4.19 / §11.4.21): HelixConstitution prescribes a project-wide work-tracker file structure. Lava's equivalent is docs/CONTINUATION.md (single-file source-of-truth handoff per §6.S) plus per-issue closure logs in .lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md (per §6.O). The substance is equivalent (status + type + history); the file structure differs. Lava does not plan to migrate to the HelixConstitution file structure unless / until the operator commissions that work; the equivalence is documented in §6.AD.3.

(2026-06-14 correction: **Build-resource stats tracker** (§11.4.24 / CM-BUILD-RESOURCE-STATS-TRACKER) is now WIRED — scripts/build-stats-sample.sh + scripts/build-stats-report.sh are present + executable; see the gate-inventory row above. **Subagent-delegation audit** (§11.4.20 / CM-SUBAGENT-DELEGATION-AUDIT) now has a scanner — scripts/check-subagent-delegation-audit.sh, verify-all-wired. **Script documentation sync** (§11.4.18 / CM-SCRIPT-DOCS-SYNC) now has BOTH the verify-all scanner scripts/check-script-docs-sync.sh AND pre-push Check 9 at .github/hooks/pre-push:195-212. The three entries below are RETAINED only as historical context; their current status is in the inventory.)

- **Issues / Issues\_Summary / Fixed / Fixed\_Summary trackers** remain the only genuinely paper-only / equivalence-mapped class — they are intentionally mapped to .lava-ci-evidence/ ledgers per §6.AD.3 Path B (CLOSED-BY-EQUIVALENCE), not pending work.

## How to read this document

- **wired**: a script enforces the rule; pre-push or similar gate fires on violation; can demonstrate by running the falsifiability rehearsal cited in the original closure commit
- **paper-only**: the rule is documented in CLAUDE.md / AGENTS.md / Constitution.md but no mechanical gate enforces it; operator + reviewer manually verify
- **blocked**: the rule cannot be enforced today due to missing infrastructure; closure work is tracked under §6.AD-debt

# How to wire a new gate

1. Read the corresponding Constitution.md §11.4.X clause for the rule's intent
2. Pick the enforcement surface: pre-push hook (.githooks/pre-push), constitution check (scripts/check-constitution.sh), distribute script (scripts/firebase-distribute.sh), tag script (scripts/tag.sh)
3. Add the gate logic; include `|| true` on every `grep -oE` pipe to avoid `set -e` killing the gate on zero matches
4. Add a falsifiability rehearsal: deliberately break a target file, re-run the gate, confirm the violation message includes the target path
5. Add a hermetic test under tests/<area>/ exercising both the positive and negative paths
6. Update this document — flip      to      for the gate's row
7. Commit + push via scripts/commit\_all.sh with proper Bluff-Audit stamp + Classification line

## Cross-references

- Lava CLAUDE.md §6.AD (HelixConstitution Inheritance Mandate) + §6.AD-debt
- HelixConstitution Constitution.md §11.4.1 through §11.4.24
- HelixConstitution CLAUDE.md (universal CLAUDE.md inherited by every consuming project)